

Netflix Dataset - Exploratory Data Analysis (EDA)

Introduction

Exploratory Data Analysis (EDA) is an important step in understanding a dataset before performing advanced analysis or modeling. In this project, the Netflix Titles Dataset was analyzed to identify trends, patterns, and insights related to Netflix content.

The analysis focused on content distribution, release trends, ratings, and country-wise contributions using statistical methods and visualizations.

Objective

The main objectives of this project were:

- Explore the Netflix dataset and understand its structure.
- Clean and prepare the data for analysis.
- Identify trends and patterns in Netflix content.
- Visualize important findings using graphs and charts.
- Generate meaningful insights from the data.

Dataset

The Netflix Titles Dataset contains information about movies and TV shows available on Netflix.

Features Included

- Title
- Type (Movie / TV Show)
- Director
- Cast
- Country
- Date Added
- Release Year
- Rating
- Duration
- Genre

Methodology

1. Data Loading and Exploration

The dataset was loaded using Pandas and basic information such as columns, data types, and dataset size was examined.

2. Data Cleaning

The dataset was checked for missing values and duplicate records. Necessary cleaning was performed to improve data quality.

3. Summary Statistics

Basic statistical analysis was carried out using functions such as `info()` and `describe()` to better understand the dataset.

4. Correlation Analysis

A correlation heatmap was created to observe relationships between numerical features.

5. Data Visualization

Several charts and graphs were created to analyze:

- Movies vs TV Shows distribution
- Content release trends by year
- Top content-producing countries
- Content ratings distribution
- Monthly content additions

Results and Insights

The analysis provided the following key insights:

- Movies are available in much larger numbers than TV Shows.
- Netflix content increased significantly after 2015.
- The United States contributes the highest amount of content.
- TV-MA is one of the most common content ratings.
- Visualizations helped identify important trends in content growth and distribution.

Tools and Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn

Conclusion

This project demonstrated how Exploratory Data Analysis can be used to understand large datasets and uncover meaningful insights. Through data cleaning, statistical analysis, and visualization, important patterns in Netflix content were identified, helping to better understand content trends on the platform.